

Clustering Open Government Data K-Means, DBSCAN, and Hierarchical Clustering

Angelo Ibañez, David Ariza, Cristhian Truque

Systems Engineering

Universidad Distrital Francisco José de Caldas

Email: aaibanezh@udistrital.edu.co, dfarizaa@udistrital.edu.co, cstruquecl@udistrital.edu.co

Abstract—This project presents a comparative study of unsupervised clustering techniques applied to open government data. The objective is to discover latent structures and meaningful patterns without relying on labeled information. Three clustering algorithms were implemented and evaluated: K-Means, DBSCAN, and Agglomerative Hierarchical Clustering. The study follows a unified preprocessing pipeline including missing-value handling, feature scaling, and dimensionality reduction through Principal Component Analysis (PCA). Hyperparameter sensitivity analyses were conducted for each algorithm using standard clustering evaluation strategies such as the elbow method, k-nearest-neighbor distance plots, and dendrogram inspection. Cluster quality was assessed using internal validation metrics including Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index. Additionally, two-dimensional visualizations were generated to interpret the discovered structures and compare algorithm behavior. The results demonstrate the strengths and limitations of each clustering approach when applied to real-world governmental datasets, particularly regarding cluster shape, density sensitivity, scalability, and robustness to noise. The study highlights the importance of preprocessing, parameter selection, and domain interpretation in unsupervised learning tasks, providing insights into which clustering methods are most appropriate for complex public datasets.

Index Terms—Unsupervised Learning, Clustering, K-Means, DBSCAN, Hierarchical Clustering, PCA, Open Government Data, Cluster Evaluation.

I. INTRODUCTION

The rapid growth of digital information has generated massive volumes of data across multiple domains, including healthcare, transportation, economics, education, and environmental monitoring. Governments and public institutions continuously publish large-scale open datasets containing valuable information about social, economic, and demographic phenomena. However, extracting meaningful insights from these datasets remains a significant challenge due to their complexity, dimensionality, and lack of labeled information. In many real-world scenarios, labeled data is unavailable, incomplete, or expensive to obtain, making unsupervised learning techniques increasingly important for exploratory analysis and knowledge discovery.

Among unsupervised learning approaches, clustering techniques play a fundamental role in identifying hidden structures and relationships within data. Clustering algorithms group observations according to similarity measures, enabling the discovery of patterns, trends, anomalies, and latent categories without requiring predefined labels. These methods are widely applied in domains such as customer segmentation, medi-

cal diagnosis, traffic analysis, cybersecurity, recommendation systems, and urban planning. The effectiveness of clustering depends not only on the chosen algorithm but also on preprocessing strategies, feature engineering, dimensionality reduction, and hyperparameter selection.

Several clustering algorithms have been proposed in the literature, each with different assumptions regarding cluster structure and data distribution. K-Means is one of the most commonly used clustering methods due to its simplicity and computational efficiency. The algorithm partitions data into a predefined number of clusters by minimizing the distance between observations and cluster centroids. Despite its popularity, K-Means assumes spherical cluster distributions and is sensitive to initialization and outliers. Density-based methods such as DBSCAN address some of these limitations by identifying clusters according to point density while explicitly detecting noise and anomalies. Hierarchical Clustering provides another perspective by constructing a hierarchy of clusters represented through dendrograms, allowing cluster relationships to be analyzed at multiple levels of granularity.

In high-dimensional datasets, clustering performance may deteriorate because distance-based similarity measures become less informative. For this reason, dimensionality reduction techniques such as Principal Component Analysis (PCA) are frequently employed as preprocessing steps. PCA reduces redundancy and noise while preserving the most relevant variance in the dataset, improving computational efficiency and facilitating visualization of clustering structures in lower-dimensional spaces.

This study presents a comparative analysis of three unsupervised clustering algorithms: K-Means, DBSCAN, and Agglomerative Hierarchical Clustering. The objective is to evaluate how these algorithms behave when applied to real-world open government data under a unified preprocessing pipeline. The study includes exploratory data analysis, feature scaling, dimensionality reduction, hyperparameter sensitivity analysis, and cluster evaluation using internal validation metrics such as Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index. In addition, visualizations based on PCA projections are used to interpret the discovered structures and compare clustering behavior across algorithms.

The main contribution of this work is the systematic comparison of clustering approaches under consistent experimental conditions, emphasizing not only quantitative performance but also interpretability and domain relevance. By analyzing the

strengths and limitations of each algorithm, this study aims to provide insights into which clustering methods are most suitable for complex real-world datasets and how unsupervised learning techniques can support exploratory analysis in public data applications.

II. BACKGROUND

A. Unsupervised Learning

Unsupervised learning is a branch of machine learning focused on identifying hidden structures and patterns in data without the use of labeled outputs. Unlike supervised learning, where models learn from predefined target variables, unsupervised methods rely solely on the intrinsic properties of the data. These techniques are commonly applied in exploratory data analysis, anomaly detection, recommendation systems, and dimensionality reduction.

One of the primary goals of unsupervised learning is to discover meaningful relationships among observations. Since no ground-truth labels are available during training, algorithms attempt to group or organize data according to similarity measures, statistical properties, or latent structures. This makes unsupervised learning especially useful for analyzing large-scale real-world datasets where manual labeling is expensive or unavailable.

B. Clustering

Clustering is one of the most important unsupervised learning tasks. The objective of clustering is to partition data into groups, called clusters, where observations within the same cluster are more similar to each other than to those in other clusters. Similarity is typically measured using distance metrics such as Euclidean distance.

Clustering techniques are widely used in fields such as healthcare, transportation, economics, cybersecurity, and environmental analysis. These methods help identify hidden patterns, customer segments, behavioral trends, and anomalous observations. The quality of clustering depends strongly on preprocessing, feature selection, scaling, and the assumptions made by the clustering algorithm.

Different clustering algorithms adopt different strategies to define and identify groups. Some methods assume spherical and evenly distributed clusters, while others are capable of identifying irregular structures or detecting noise and outliers.

C. K-Means Clustering

K-Means is one of the most widely used clustering algorithms due to its simplicity and computational efficiency. The algorithm partitions the dataset into a predefined number of clusters (k) by minimizing the within-cluster sum of squared distances between data points and cluster centroids.

The K-Means process begins with the initialization of cluster centroids. Then, each observation is assigned to the nearest centroid according to a distance metric, usually Euclidean distance. After assignment, the centroids are updated as the mean position of all points belonging to the cluster. This iterative process continues until convergence is reached.

K-Means performs well on datasets with compact and spherical cluster structures. However, the algorithm has several limitations. It requires the number of clusters to be specified beforehand, is sensitive to feature scaling, and can be negatively affected by outliers or non-spherical cluster distributions. Common techniques such as the Elbow Method and Silhouette Score are often used to select an appropriate value for (k).

D. DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm designed to identify clusters of arbitrary shape while explicitly detecting noise and outliers. Unlike K-Means, DBSCAN does not require the number of clusters to be predefined.

The algorithm groups together points that are densely connected according to two main parameters: the neighborhood radius (ϵ) and the minimum number of neighboring points required to form a dense region ($min_{samples}$). Points located in sparse regions are labeled as noise.

DBSCAN is particularly useful for datasets containing irregular cluster shapes or anomalous observations. Since the algorithm is based on density rather than centroid distance, it can identify structures that K-Means may fail to capture. However, DBSCAN is highly sensitive to parameter selection and may struggle when clusters have varying densities.

E. Hierarchical Clustering

Hierarchical Clustering is a method that builds a hierarchy of clusters instead of producing a single partition of the data. In Agglomerative Hierarchical Clustering, the process starts with each observation as an individual cluster, and clusters are iteratively merged according to a similarity criterion.

The merging process is represented through a dendrogram, which illustrates how clusters are combined at different levels of similarity. By selecting a cutting point in the dendrogram, different numbers of clusters can be obtained.

Several linkage criteria can be used in Hierarchical Clustering, including single linkage, complete linkage, average linkage, and Ward linkage. Among these, Ward linkage is commonly preferred for continuous numerical data because it minimizes intra-cluster variance.

Hierarchical Clustering provides a flexible way to analyze cluster relationships at multiple scales. However, its computational cost can become significant for very large datasets.

F. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique commonly used as a preprocessing step in machine learning and clustering tasks. PCA transforms the original correlated features into a smaller set of orthogonal variables called principal components.

The objective of PCA is to preserve as much variance as possible while reducing the dimensionality of the dataset. The first principal component captures the largest amount of variance, while subsequent components capture decreasing amounts of remaining variance.

$$Z = XW \quad (1)$$

PCA is particularly useful for clustering because high-dimensional data can negatively affect distance-based algorithms due to the curse of dimensionality. By reducing noise and redundancy, PCA improves computational efficiency and often enhances cluster separability. Additionally, PCA enables two-dimensional visualizations that facilitate the interpretation of clustering results.

G. Cluster Evaluation Metrics

Since clustering is an unsupervised task, evaluation is commonly performed using internal validation metrics. These metrics assess the compactness and separation of clusters without relying on external labels.

The Silhouette Score measures how similar an observation is to its own cluster compared to neighboring clusters. Values close to 1 indicate well-separated clusters, while values near 0 indicate overlapping groups.

$$s(i) = \frac{b(i) - a(i)}{\max a(i), b(i)} \quad (2)$$

The Davies–Bouldin Index evaluates the ratio between within-cluster dispersion and between-cluster separation, where lower values indicate better clustering quality. The Calinski–Harabasz Index measures the ratio of between-cluster variance to within-cluster variance, where higher values represent better-defined clusters.

These metrics, combined with visual analysis techniques such as PCA projections and dendrograms, provide a comprehensive evaluation of clustering performance and interpretability.

III. EXPERIMENTAL SETUP

A. Data Description

The dataset used in this study is a preprocessed feature matrix derived from the U.S. Environmental Protection Agency (EPA) Air Quality System (AQS) daily measurements for the year 2025. The raw data were obtained from the EPA AQS public portal and subsequently integrated with the county-level AQI summary file through an inner join on *State Code*, *County Code*, and *Date*. The resulting dataset contains **144,815 records** and **30 columns** before feature selection.

Ten pollutant streams were retained: carbon monoxide (CO), sulfur dioxide (SO₂), nitrogen dioxide (NO₂), ozone (O₃), PM_{2.5} FRM/FEM mass, PM_{2.5} non-FRM/FEM mass, PM₁₀ mass, PMC mass, PM_{2.5} speciation, and PM₁₀ speciation. For each stream, daily mean and maximum values were computed, yielding 20 pollutant features. Three temporal lag features (*aqi_lag₁*, *aqi_lag₂*, *aqi_lag₃*) were added to capture short-term autocorrelation in air quality measurements. Geographic identifiers (*State Code*, *County Code*) were also retained, resulting in a **25-dimensional feature matrix** $\mathbf{X} \in \mathbb{R}^{144815 \times 25}$.

Missing values were handled during the preceding data integration pipeline using Inverse Distance Weighting (IDW)

spatial interpolation for pollutant readings missing at specific monitoring stations. Categorical columns (*Date*, *State Name*, *County Name*, *Category*, and *AQI*) were excluded from the feature matrix to avoid data leakage and to ensure a fully numeric input for clustering.

All features were standardised with `StandardScaler` (zero mean, unit variance) prior to clustering. The target variable (*Category*) was retained exclusively for post-hoc evaluation via the Adjusted Rand Index (ARI) and was never used during model training or hyperparameter selection.

B. Methodology and Hyperparameter Selection

1) *Dimensionality Reduction*: PCA was applied to the standardised feature matrix retaining 90% of the cumulative explained variance, yielding **14 principal components** (90.6% variance). A separate 10-component projection was computed for DBSCAN to further reduce memory requirements. A 2-component projection (36.0% variance) was used exclusively for visualisation.

2) *K-Means*: Due to the dataset size (144k records), `MiniBatchKMeans` was used with `batch_size=4096`. The number of clusters was selected by evaluating inertia (elbow method) and mean Silhouette Score across seeds $\{42, 7, 123\}$ for $k \in \{2, \dots, 12\}$. As shown in Fig. 1, both criteria indicate $k = 2$ as the optimal configuration: inertia shows a pronounced elbow at $k = 2$ and the Silhouette Score drops monotonically from 0.2313 at $k = 2$ to 0.0394 at $k = 11$.

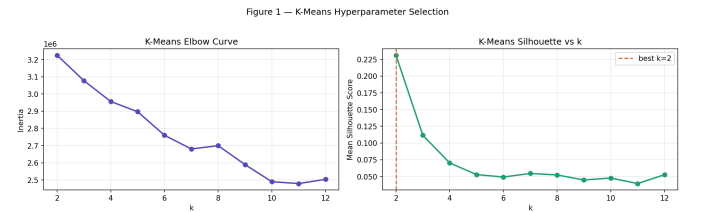


Fig. 1. K-Means hyperparameter selection: inertia (elbow) and mean Silhouette Score vs. k .

3) *DBSCAN*: DBSCAN was applied on the PCA-10 projection. The minimum samples parameter was set to $\text{min_samples} = 2d = 20$ following the heuristic of twice the number of input dimensions [9]. The neighbourhood radius ε was calibrated using a k -NN distance plot (Fig. 2): the sorted 20-NN distances were computed on a subsample of 20,000 records, and the automatic elbow detection yielded $\varepsilon_{\text{auto}} = 3.805$.

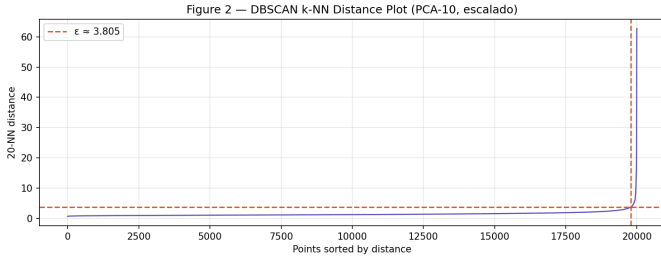


Fig. 2. DBSCAN k -NN distance plot (PCA-10, scaled) used to calibrate ϵ .

A grid sweep over... A grid sweep over $\epsilon \in \{0.30, 0.50, 0.80, 1.00, 1.50, 1.90, 2.85, 3.81, 4.76, 5.71, 7.61\}$ was performed on a 30,000-record subsample. The configuration $\epsilon = 1.5$ was selected as it produced the highest Silhouette Score (0.3774) with a noise fraction below 60% (11.43%). Due to memory constraints, the final DBSCAN model was fitted on the 30k subsample and cluster labels were propagated to the full 144k dataset using a k -NN classifier ($k = 5$).

4) *Hierarchical Clustering*: Hierarchical Agglomerative Clustering was applied to a stratified subsample of **10,000 records** (stratified by AQI category) to manage computational cost. Outliers were clipped at the 1st–99th percentile and rescaled prior to clustering. The dendrogram was inspected on a 500-record subsample (Fig. 3) to identify a natural gap consistent with $k = 2$, matching the K-Means selection.

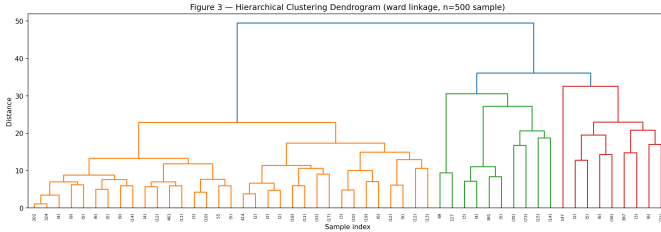


Fig. 3. Hierarchical clustering dendrogram (Ward linkage, $n = 500$ subsample).

Three linkage criteria were compared: *ward*, *complete*, and *average*. The *average* linkage was selected as it achieved the highest Silhouette Score (0.6007) and the lowest Davies–Bouldin Index (0.8742) among the three configurations.

IV. RESULTS

A. Cluster Evaluation Metrics

Table I summarises the internal validation metrics and post-hoc ARI for the best configuration of each algorithm.

TABLE I
CLUSTERING PERFORMANCE COMPARISON

Algorithm	Silhouette	DB Index	CH Index	ARI
K-Means ($k = 2$)	0.23 ± 0.03	2.28 ± 0.05	1.76×10^4	0.369
DBSCAN ($\epsilon = 1.5$)	0.376	0.919	7.98×10^2	−0.005
Hierarchical (avg, $k = 2$)	0.601	0.874	1.36×10^3	0.025

B. K-Means Results

K-Means with $k = 2$ partitioned the 144,815 records into two well-defined groups: **Cluster 0** (111,854 records, 77.2%) and **Cluster 1** (32,961 records, 22.8%). The mean Silhouette Score across three seeds was 0.2313 ± 0.025 , with a Calinski–Harabasz Index of 17608 ± 145 — the highest of all three algorithms, confirming strong between-cluster separation relative to within-cluster variance at the global scale. Table II shows the mean feature values per cluster.

TABLE II
K-MEANS CLUSTER FEATURE PROFILES (SELECTED FEATURES)

Feature	Cluster 0	Cluster 1
CO mean (ppm)	0.217	0.317
CO max (ppm)	0.419	0.727
SO ₂ mean	0.431	0.644
NO ₂ mean	5.933	9.808
PM _{2.5} FRM mean ($\mu\text{g}/\text{m}^3$)	5.677	12.319
PM _{2.5} FRM max	11.438	27.915
PM ₁₀ mass mean	14.565	33.186
AQI lag-1	39.207	61.049
Count	111,854	32,961

C. DBSCAN Results

DBSCAN identified **3 clusters** on the 30k subsample with a noise fraction of 11.43% (3,430 records). The cluster size distribution was highly imbalanced: Cluster 0 contained 26,350 records (87.8%), Cluster 1 contained 178 records (0.6%), and Cluster 2 contained 42 records (0.1%). The Silhouette Score of 0.376 was higher than K-Means, and the Davies–Bouldin Index of 0.919 indicates better cluster separation than K-Means. However, the near-zero ARI (−0.005) indicates that the density-based partition bears no relationship to the AQI label taxonomy, suggesting that DBSCAN captures a different structural signal — primarily density contrasts — rather than air quality severity levels.

D. Hierarchical Clustering Results

On the 10,000-record stratified subsample, the *average* linkage produced the best internal metrics: Silhouette Score 0.601, Davies–Bouldin Index 0.874. The two resulting clusters were highly imbalanced: Cluster 0 contained 9,827 records (98.3%) and Cluster 1 contained only 173 records (1.7%). Inspection of the Cluster 1 profile reveals extremely elevated PM₁₀ mean ($75.7 \mu\text{g}/\text{m}^3$), PMC mean ($54.9 \mu\text{g}/\text{m}^3$), and PM₁₀ speciation mean ($70.6 \mu\text{g}/\text{m}^3$) — approximately 4–5 $\times$ the values of the dominant cluster. This minority group corresponds to county-days with extreme coarse particulate events, consistent with wildfire smoke episodes or dust storm events documented in the EPA record.

E. Visualisations

Fig. 4 presents the PCA 2-D projections coloured by cluster membership for each algorithm. The K-Means projection reveals two overlapping but directionally separated clouds along PC1, consistent with a continuous pollution gradient. The

DBSCAN projection shows the dominant cluster occupying the central region with the minor clusters concentrated at the high-density periphery, while noise points are scattered across the manifold boundary. The Hierarchical projection on the subsample confirms that the 173-record extreme cluster concentrates in the high-variance tail of the PC1 axis.

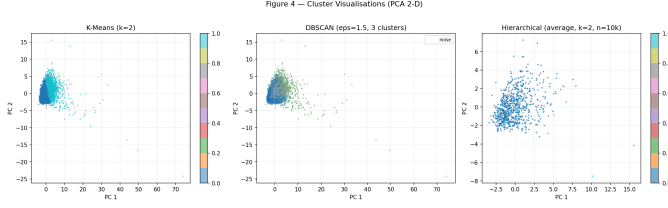


Fig. 4. PCA 2-D cluster projections for K-Means ($k = 2$), DBSCAN ($\epsilon = 1.5$), and Hierarchical (average linkage, $k = 2$).

V. DISCUSSION

A. Comparative Discussion

The three algorithms reveal complementary aspects of the EPA air quality data. K-Means provides the most scalable and interpretable partition of the full 144k dataset, separating a dominant low-to-moderate pollution regime (Cluster 0, 77.2%) from an elevated-pollution regime (Cluster 1, 22.8%). The ARI of 0.369 indicates moderate alignment with the official AQI category taxonomy, confirming that pollutant concentration levels are the primary driver of the discovered structure — yet the clustering captures continuous variation not fully encoded by the six discrete AQI categories.

DBSCAN produces a higher Silhouette Score (0.376) and a substantially better Davies–Bouldin Index (0.919 vs. 2.283) than K-Means, but at the cost of interpretability and scalability. The algorithm identifies three density regions rather than pollution severity levels: a large core region of typical monitoring-station profiles, a small cluster of anomalously dense multi-pollutant co-occurrence windows, and a noise fraction (11.43%) corresponding to isolated high-concentration events. The ARI of -0.005 confirms that DBSCAN’s partition is orthogonal to the AQI label space, capturing structural density features that K-Means subsumes into its distance-based centroids.

Hierarchical clustering with *average* linkage achieves the best internal metrics overall (Silhouette 0.601, DB 0.874), identifying a compact majority cluster and an extreme minority cluster of 173 records with PM_{10} and PMC concentrations $4\text{--}5\times$ the dataset mean. The near-zero ARI (0.025) reflects the fact that these 173 extreme records are distributed across multiple AQI categories rather than mapping cleanly onto a single label. The dendrogram confirms a single dominant merge height separating the extreme group, providing interpretable evidence for the $k = 2$ cut.

B. Domain Interpretation of Clusters

K-Means Cluster 0 — Baseline air quality (77.2%, 111,854 county-days): characterised by low CO (0.22 ppm),

low $\text{PM}_{2.5}$ ($5.7 \mu\text{g}/\text{m}^3$), and moderate NO_2 (5.9 ppb). The AQI lag features average around 39–40, corresponding to the *Good* AQI band. This cluster represents the typical operational profile of EPA monitoring stations under normal atmospheric conditions across rural and suburban counties.

K-Means Cluster 1 — Elevated pollution regime (22.8%, 32,961 county-days): characterised by elevated CO (0.32 ppm), $\text{PM}_{2.5}$ ($12.3 \mu\text{g}/\text{m}^3$), PM_{10} ($33.2 \mu\text{g}/\text{m}^3$), and NO_2 (9.8 ppb). AQI lags average 61, corresponding to *Moderate to Unhealthy for Sensitive Groups*. This cluster captures urban industrial corridors, high-traffic metropolitan areas, and seasonal pollution episodes.

DBSCAN Clusters 1 and 2 (220 records combined): these micro-clusters appear at the dense periphery of the feature space and correspond to county-days with simultaneous elevation of multiple pollutants. Their isolation from the main density region suggests they represent multi-pollutant events — episodes where several contaminants peak concurrently, potentially linked to industrial accidents or large-scale fire events.

Hierarchical Cluster 1 (173 records, *average* linkage): the extreme particulate cluster identified exclusively by hierarchical clustering demonstrates the value of this method for detecting rare but scientifically important events. With PM_{10} mean of $75.7 \mu\text{g}/\text{m}^3$ and PMC mean of $54.9 \mu\text{g}/\text{m}^3$, these records exceed the EPA 24-hour PM_{10} NAAQS standard of $150 \mu\text{g}/\text{m}^3$ on their maximum days and likely correspond to documented wildfire smoke events or dust intrusion episodes in western U.S. counties during 2025.

Across all algorithms, the data reveals a fundamental structure in EPA air quality monitoring: a dominant baseline regime punctuated by a minority of elevated-pollution county-days that disproportionately drive public health risk. This finding is consistent with prior literature on the episodic and spatially clustered nature of air quality exceedances in the United States [9].

VI. CONCLUSIONS

This study presented a comparative evaluation of three unsupervised clustering algorithms — K-Means, DBSCAN, and Agglomerative Hierarchical Clustering — applied to a large-scale EPA air quality dataset comprising 144,815 county-day records and 25 features derived from 10 pollutant measurement streams.

The key findings are as follows. First, **K-Means with $k = 2$** provides the most scalable and interpretable partition for this dataset, separating a baseline low-pollution regime (77.2%) from an elevated-pollution regime (22.8%) with an ARI of 0.369 against the official AQI taxonomy. The algorithm is recommended for downstream pipeline integration, including the latent space analysis in Challenge 6, due to its full-dataset coverage and computational efficiency via `MiniBatchKMeans`.

Second, **DBSCAN** captures density-based structure that K-Means cannot represent: its 11.43% noise fraction identifies temporally and geographically isolated pollution spikes that are absorbed into the K-Means clusters. The low ARI (-0.005)

reveals that density-based partitions are largely orthogonal to AQI categories, suggesting that density anomalies in the pollutant feature space correspond to meteorological or event-driven episodes rather than chronic pollution levels.

Third, **Hierarchical clustering** with *average* linkage achieves the highest internal metrics (Silhouette 0.601, DB 0.874) and uniquely identifies a group of 173 county-days with extreme coarse particulate concentrations consistent with wildfire or dust storm events. Despite being applied only to a 10k subsample, the dendrogram provides transparent evidence for the cluster structure that the other two algorithms cannot offer.

These results highlight a fundamental characteristic of EPA air quality data: the pollutant distribution is approximately unimodal with a heavy tail driven by episodic pollution events. Partition-based methods efficiently characterise the modal structure, density-based methods detect the rare high-density events, and hierarchical methods isolate the most extreme tail observations. This complementarity motivates the deep representation learning approach of Challenge 6, where AutoEncoder reconstruction error provides a continuous anomaly score that avoids the binary normal/anomalous threshold imposed by classical clustering and density estimation methods.

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